Training parts versus matching parts in pose-based gait recognition

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Abstract

Part-aware pose-based gait recognizers usually combine two interventions: identity objectives on body regions during training and fusion of regional similarities during retrieval. We separate them on one fixed multistream graph encoder for two-dimensional poses and then test the result on GP-Gait. On CASIA-B development identities, seven training objectives, including global and part supervised contrastive (SupCon) terms and cross-entropy controls, were each trained with five seeds and evaluated under synthetic missing joints. No auxiliary contrastive objective differed reliably from a metric-learning baseline; an initial three-seed part-SupCon penalty (-6.12 Rank-1 points, 95% interval $[-10.29, -2.00]$) shrank to -1.27 $[-5.24, 2.74]$. Fusing normalized raw part pools from frozen checkpoints instead improved corrupted-probe Rank-1 under every objective ($+2.32$ to $+4.28$ points), and controls located the gain in score fusion rather than in learned heads or confidence weighting. On Gait3D test identities, evaluated once with a frozen

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recipe, the matcher raised Rank-1 from 21.88% to 23.86% (+1.99 [0.61, 3.38]), level with GPGait reproduced on identical data (23.13%). The gain did not transfer to GPGait itself: its trained part branches added 9.08 points on CASIA-B and 5.17 on Gait3D, whereas adding raw parts lowered accuracy by 2.40 and 3.17 points. Our encoder also remains far below GPGait on CASIA-B (54.04% versus 80.91%). Whether local supervision or local matching carries a part-aware gain therefore depends on the encoder, and both should be measured on the same frozen checkpoints.

Keywords: Gait recognition, Pose-based biometrics, Occlusion robustness, Supervised contrastive learning, Part-based matching, Controlled evaluation, Negative results

1. Introduction

Pose-based gait recognition retrieves identities from joint trajectories. The representation is compact and depends less on clothing, background and illumination than silhouettes do [1], and graph encoders such as ST-GCN, GaitGraph and GaitGraph2 make its spatial and temporal structure explicit [2–4]. Recognition is nevertheless limited by what the pose estimator recovers. Occlusion and detector failures remove joints or produce wrong tracks, so partial body evidence is a central difficulty [5, 6].

Body regions offer two distinct interventions. During training, a local identity objective can push each region to discriminate individuals. During retrieval, regional similarities can supplement a global sequence score. Part-aware systems usually adopt both at once, yet a region can contribute useful complementary matching evidence without being independently dis-

criminative under a local loss. Supervised contrastive (SupCon) learning [7] offers a natural local objective, but whether it helps gait parts is an empirical question. Attributing an improvement to the intended mechanism requires separating the two interventions.

We perform that separation on one fixed multistream graph encoder whose input representation, augmentation, batches, base losses, checkpoint selection and evaluation masks are shared; only the auxiliary objective or the retrieval descriptor changes. No auxiliary global or part SupCon term reliably improves a metric-learning baseline, and a part-SupCon penalty seen in an initial three-seed factorial is much smaller with five seeds. Matching normalized raw part pools from the same frozen checkpoints improves corrupted-probe accuracy under every objective and on both CASIA-B and Gait3D, including Gait3D test identities evaluated once with a recipe frozen beforehand. Weighting parts by detector confidence adds nothing over uniform averaging. The result does not transfer to GPGait [8], whose part-specific branches are trained with their own losses: there the trained parts add 5 to 9 points and raw part matching lowers accuracy (Fig. 1). The contributions are:

1. A controlled comparison of global and part SupCon, cross-entropy controls and a metric-learning baseline on one pose-gait encoder, with a five-seed replication that revises a three-seed finding.
2. Checkpoint-only controls (raw pools, uniform weights, ten random heads, five fusion weights) and raw matching for every trained objective, which locate the retrieval gain in regional score fusion.
3. An external Gait3D evaluation with a recipe frozen before test access,

a detector-confidence stratification, and GPGait and GaitGraph2 reproduced on identical data.

4. A transfer test on GPGait showing that its trained part branches, not training-free regional matching, carry its part-aware gain.

Which part intervention helps therefore depends on the encoder, and the design check we recommend is to compare global, raw regional and trained regional matching on the same frozen checkpoints before crediting either one.

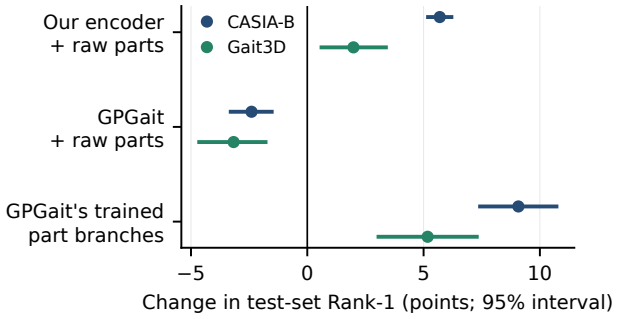


Figure 1: Test-set Rank-1 change from regional evidence. Training-free raw part matching helps our encoder on CASIA-B (standard protocol, five seeds, paired t interval) and Gait3D (five seeds, crossed bootstrap) but lowers the accuracy of GPGait, whose trained part branches carry a large gain instead (one model per dataset; intervals resample test identities).

2. Related work

Contrastive and metric objectives. Contrastive objectives depend on positive-pair construction and projection geometry [7, 9, 10], which can be described by alignment and uniformity [11]; other self-supervised objectives avoid negatives or regularize variance and redundancy [12–15]. Skeleton-specific con-

trastive learning serves action recognition and re-identification [16–18] and large-scale gait pretraining [19]. Sampling and pair weighting shape metric objectives [20, 21], loss comparisons require matched training and evaluation [22], and interactions between losses can be inspected through their gradients [23, 24]. None of these results determines whether an independent identity objective helps each anatomical gait region.

Pose-gait encoders and missing evidence. PoseGait uses body pose and human priors [1], GaitGraph and GaitGraph2 learn graph features [3, 4], GPGait adds part-aware local–global graph extraction [8], and SkeletonGait renders skeleton maps [25]; OpenGait and FastPoseGait standardize implementations [26, 27], and skeleton action encoders offer further graph and heatmap designs [28–32]. Real occlusion combines missing joints with mislocalization [5, 6, 33], and GREW, Gait3D and SUSTech1K supply unconstrained or occluded conditions [34–36].

Part-aware representations. GaitSet pools silhouette sets, and disentangling methods separate identity from appearance or covariates [37–39]. GaitPart, GaitGL and 3D local CNNs extract complementary local and global features [40–42]; strip-based, context-sensitive temporal, dynamic-region and template representations extend the aggregation choices [43–46]. These systems train and match local features together. We claim no novelty for anatomical pooling or global–local fusion; we ask which of the two carries the gain, first with our encoder held fixed and then on GPGait.

3. Study design

Figure 2 summarizes the design.

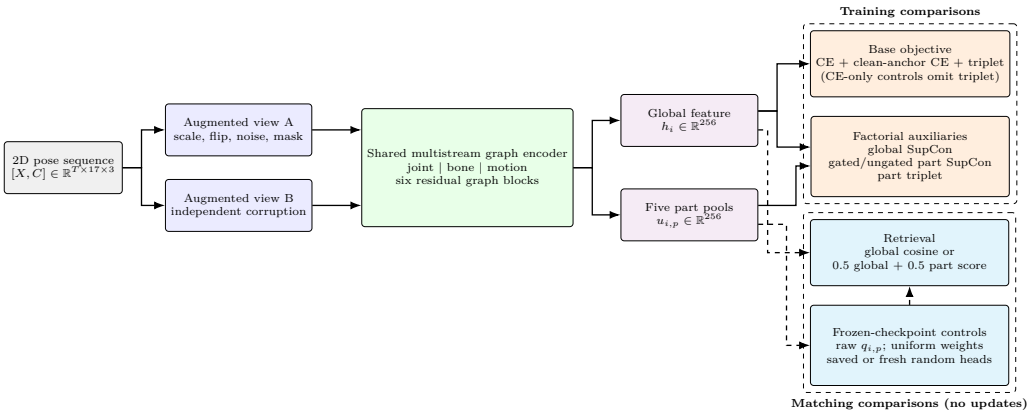


Figure 2: Separation of part-aware training from part-aware matching. Solid arrows denote the shared forward path. The upper branch contains the training objectives; the lower branch contains frozen-checkpoint descriptor controls and performs no parameter updates.

3.1. Encoder and regional descriptors

Let $X \in \mathbb{R}^{T\times 17\times 2}$ be a normalized pose and $C \in [0,1]^{T\times 17}$ its confidence. The encoder builds three input streams: joints $[X, C]$; bones relative to a rooted parent, observed only when both endpoints are present; and motion at lags one and two, observed only when both temporal endpoints are present. Stream-specific stems feed six residual graph blocks that use learned perturbations of the physical adjacency and temporal convolutions with dilations 1, 2 and 4. The width grows from 64 to 256, and confidence-weighted global pooling gives a normalized 256-dimensional embedding h_i.

Five overlapping COCO regions define local pools: the torso (shoulders and hips), two arms (shoulder, elbow, wrist) and two legs (hip, knee, ankle); face joints are excluded. Confidence-weighted pooling within part p gives $u_{i,p} \in \mathbb{R}^{256}$. The deterministic raw descriptor is $q_{i,p} = u_{i,p}/\|u_{i,p}\|_2$; the learned alternative passes $u_{i,p}$ through a part-specific two-layer head and ℓ_2

normalization to give $z_{i,p}$. Part reliability is

$$r_{i,p} = \frac{1}{T_i|p|} \sum_{t=1}^{T_i} \sum_{j \in p} C_{i,t,j}, \tag{1}$$

where T_i counts real rather than padded frames.

3.2. Training objectives

Each batch samples 16 identities with four sequences each. For matched CE, the metric baseline and every auxiliary variant, two views of each sequence receive scale, flip and coordinate-noise augmentation; each view independently stays geometrically complete with probability 0.3 and otherwise receives random-joint, body-part or temporal masking, with the maximum severity rising linearly from 0.05 to 0.5 over the first half of training. Dynamic-joint corruption is reserved for evaluation. The metric baseline uses

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{CE,aug}} + 0.5\,\mathcal{L}_{\text{CE,clean}} + \mathcal{L}_{\text{Tri}}, \tag{2}$$

where the first term classifies both augmented views, the clean anchor classifies the unaugmented sequence, and $\mathcal{L}_{\text{Tri}}$ is batch-hard triplet loss [47] on h_i with margin 0.2; same-source views are excluded as triplet positives. Global SupCon $\mathcal{L}_{\text{gSC}}$ acts on a projection of h_i with every same-identity view as positive. Part SupCon $\mathcal{L}_{\text{pSC}}$ applies the same construction to each $z_{i,p}$; in the gated form a positive pair is weighted by $r_{i,p}r_{j,p}$ and the anchor by $r_{i,p}$, and parts with reliability below 0.05 are excluded. The ungated form gives every part above that threshold unit weight. Part triplet $\mathcal{L}_{\text{pTri}}$ applies batch-hard triplet loss to every present part. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \gamma_g \mathcal{L}_{\text{gSC}} + \gamma_p \mathcal{L}_{\text{pSC}} + \gamma_{pt} \mathcal{L}_{\text{pTri}}. \tag{3}$$

Table 1 lists the configurations. Three CE controls bracket the baseline: plain CE without masking, CE with random joint dropout, and matched CE, which uses the baseline's partial views and clean anchor but no triplet term.

Table 1: Training objectives. All other settings are shared. "Five" and "three" mark the CASIA-B cohorts that contain each objective; Gait3D development uses CE, matched CE, the metric baseline and global + part SupCon.

Configuration	Triplet	γ_g	γ_p	γ_{pt}	Part gate	CASIA-B seeds
CE, no masking	–	0	0	0	–	five
Masked CE (joint dropout)	–	0	0	0	–	five
Matched CE	–	0	0	0	–	five
Metric baseline	✓	0	0	0	–	five, three
Global SupCon	✓	0.2	0	0	–	five, three
Part SupCon	✓	0	0.2	0	reliability	five, three
Global + part SupCon	✓	0.2	0.2	0	reliability	five, three
Global + part SupCon, ungated	✓	0.2	0.2	0	presence	three
Part triplet	✓	0	0	0.2	–	three

3.3. Retrieval descriptors and checkpoint-only controls

The global descriptor scores $s_g(i,j) = h_i^\top h_j$. A part descriptor keeps this term and averages local cosine scores over jointly reliable parts. With $v_{i,p}$ either the learned projection $z_{i,p}$ or the raw pool $q_{i,p}$,

$$s_p(i,j) = \frac{\sum_p r_{i,p} r_{j,p}\, v_{i,p}^\top v_{j,p}}{\max\left(\sum_p r_{i,p} r_{j,p}, 10^{-8}\right)}, \qquad s(i,j) = (1-\lambda)\, s_g(i,j) + \lambda\, s_p(i,j), \quad (4)$$

with $\lambda = 0.5$ declared before any comparison. Uniform matching replaces $r_{i,p}$ by the presence indicator $\mathbf{1}[r_{i,p} > 0]$, and matching falls back to s_g when no part is jointly present. In the metric baseline and global SupCon runs the part heads receive no loss, so their learned projections stay at their seeded

initialization while the backbone changes. The checkpoint-only controls resolve this confound: each selected metric-baseline checkpoint is frozen and evaluated with (i) raw pools and reliability weights, (ii) raw pools and uniform weights, (iii) the saved untrained heads and (iv) ten freshly initialized heads of the same architecture, at $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$; only $\lambda = 0.5$ is primary. Raw matching is additionally evaluated for every trained objective in the five-seed CASIA-B cohort and in both Gait3D stages.

3.4. Transfer test on GPGait

GPGait has joint, bone and angle branches. In each, a global backbone and part-specific backbones with part-restricted graph connectivity feed semantic pyramid pooling, and every pooled region receives its own projection, BNNeck layer and identity losses. This gives 19 unit-normalized embeddings, three of them whole-body, and the native distance averages their per-embedding Euclidean distances. Running the trained checkpoints through FastPoseGait's own test pipeline, we keep the 19 embeddings and the per-joint features of each branch's global backbone after temporal pooling. Five descriptors are compared: all 19 embeddings (full, as published); the three whole-body embeddings; raw parts, which are our five regions mean-pooled from each branch's global-backbone joint features and ℓ_2-normalized (15 vectors, no learned parameters); and the whole-body or full descriptor fused with raw parts on the native distance at $\lambda = 0.5$. These choices were fixed before evaluation, and the full descriptor reproduces the logged test accuracy to within 0.03 points. Each GPGait model was trained once, so intervals resample test identities (20,000 draws).

4. Experimental setup

4.1. Datasets and protocols

CASIA-B. We use the HRNet-derived [48] 2D pose release of CASIA-B [49] distributed with FastPoseGait [27]. It has 124 subjects, 11 views (0–180°) and ten sequences per view (six normal, NM; two with a bag, BG; two with a coat, CL); our copy has 13,637 of the 13,640 sequences. Each frame contains 17 COCO joints [50] with two coordinates and a confidence value. For development, a fixed split seed (2026) holds out 12 of the 74 training identities (002, 006, 012, 025, 026, 032, 043, 046, 054, 059, 061, 072); the remaining 62 train the models. Normal sequences 01–04 of each held-out identity form the gallery, and normal 05–06, bag 01–02 and clothing 01–02 form 792 probes. For the official test, identities 001–074 train and identities 075–124 are evaluated with the standard single-view protocol: normal sequences 01–04 as gallery, normal 05–06, bag 01–02 and clothing 01–02 as probes, an average over the 11 gallery views, and exclusion of identical-view matches (3,300 probes). Earlier exploratory runs in this project evaluated identities 075–124, so we treat the CASIA-B test result as within-dataset transfer rather than an untouched holdout.

Gait3D. Gait3D [35] was recorded by 39 cameras in a large supermarket; its official partition assigns 3,000 identities to training and 1,000 to testing. We use its released HRNet 2D poses (COCO-17 with confidence); its silhouettes and 3D body meshes are not used. The pose archive available to us lacks 108 training and 53 test identities, so all Gait3D results, including the reproduced baselines, use the same available-pose subset: 2,892 training identities

(18,135 sequences) and 947 test identities with 947 designated probes and 5,122 gallery sequences. Frame gaps are kept as gaps rather than interpolated. For development, 300 training identities are held out with split seed 2026; one track per identity is the probe and its other tracks form the gallery. Before this study, no Gait3D test outcome had been computed in the project. In both datasets the keypoints are pose-estimator outputs, not manual annotations; the human-provided labels are subject identity and capture condition (view and walking condition for CASIA-B, camera for Gait3D), and neither dataset labels occlusion. The final runs train on all 2,892 training identities and evaluate the test probes once.

4.2. Implementation details

Coordinates are centred on the available pelvis joints, falling back to the centroid of visible joints, and normalized by sequence body scale. Confidence is an input channel and pooling weight; it never multiplies coordinates. Training uses contiguous 60-frame clips, repeated cyclically for short sequences; evaluation uses every real frame. Training lasts 160 epochs with AdamW (learning rate 10^{-3}, weight decay 10^{-4}, five warm-up epochs, cosine decay to 1%), label smoothing 0.1, global and part SupCon temperatures 0.07 and 0.1, gradient clipping at 5 and mixed precision. The three-seed factorial (seeds 11–33) and the five-seed cohort (seeds 11–55) share this recipe but not the source revision, so they are reported separately. Training and timing used NVIDIA H100 NVL multi-instance GPU slices.

Development checkpoints are selected every five epochs on held-out training identities with the score $0.75\,\mathrm{R1}_{\mathrm{clean}} + 0.25\,\overline{\mathrm{R1}}_{0.3}$ (the second term averages body-part and random-joint corruption at severity 0.3), using the global de-

scriptor for every method. Final runs disable validation and use a budget fixed beforehand. For Gait3D the rule keeps the full 160 epochs when the last epoch lies within 0.01 of the smoothed validation maximum; this held for the metric baseline (gap 0.0031) but not for plain cross-entropy (CE; gap 0.0101, suggested 110 epochs). Both were trained for the common 160 epochs, a decision recorded before any test evaluation, with descriptor, fusion weight and recipe frozen from development.

4.3. Corruption protocols

We evaluate random joint-frame removal, temporally smoothed dynamic-joint removal, whole-part removal and a contiguous temporal interruption at nominal severities $\rho \in \{0, 0.1, 0.3, 0.5, 0.7\}$ (Supplementary Fig. S2). For whole parts and smoothed dynamic masks the removed fraction of joint-frames differs from ρ (at $\rho = 0.1$ it is 18.8% and 3.2%, at $\rho = 0.7$ 70.6% and 78.7%; Supplementary Table S4). Masks are applied to normalized poses and isolate missing evidence, not pose-estimation errors. Five deterministic realizations, independent of the training seed, are averaged at each positive severity. Only probes are corrupted in the clean-gallery protocol; in the occluded-gallery protocol gallery and probe receive independently seeded corruption of the same family and severity.

4.4. Metrics and statistics

The primary measure is Rank-1. "Clean" denotes uncorrupted probes; "corrupted probe" averages the four families at the four positive severities with a clean gallery; "corrupted gallery" averages the same grid with corrupted galleries. On CASIA-B development data we also report the single-

view NM/BG/CL average. Tables report means and sample standard deviations over seeds. Paired differences average the selected cells within each probe, pair candidate and reference by seed and identity, and resample seeds and identities as crossed factors (100,000 draws for the three-seed factorial, 20,000 otherwise); for the CASIA-B official standard protocol, where only seed-level outcomes were saved, we use a paired t interval over the five seeds. Intervals are pointwise 95% intervals without multiplicity adjustment and quantify uncertainty across many exploratory endpoints. A pre-specified development criterion for adopting a part descriptor required a clean change of at least -0.5 points, a corrupted-probe gain of at least 2 points and improvement in at least three families.

4.5. Reproduced baselines

GPGait [8] and GaitGraph2 [4] were trained with the official FastPoseGait configurations [27], changing only data paths and disabling intermediate testing. The CASIA-B runs use the standard 74/50 identity split; the Gait3D runs use exactly the available-pose subset described above, exported from the same data file. Each framework is scored with its own evaluator on the same probe and gallery partition, and each baseline was trained once.

5. Results

5.1. Auxiliary objectives do not improve the metric baseline

Table 2 reports the five-seed CASIA-B cohort and Table 3 the paired differences from the metric baseline. Two contrasts dominate. Training on partial views matters: CE without masking reaches the baseline's clean accuracy

but loses 27.77 points on corrupted probes, interval $[-33.21, -22.29]$, and masked CE loses 9.19 clean points, $[-14.67, -4.34]$. Given matched partial views, the triplet term is not resolved on CASIA-B (matched CE minus baseline: -2.26, $[-7.08, 2.86]$), but it is on Gait3D, where matched CE loses 4.33 clean and 2.46 corrupted-probe points, $[-8.22, -0.11]$ and $[-4.35, -0.39]$.

None of the auxiliary SupCon objectives is distinguishable from the metric baseline: on CASIA-B, global SupCon changes corrupted-probe accuracy by $+1.37$ points, $[-2.19, 5.02]$, part SupCon by -1.27, $[-5.24, 2.74]$, and global + part SupCon by $+1.21$, $[-2.54, 4.40]$; on Gait3D, global + part SupCon changes it by $+0.07$, $[-1.73, 1.97]$. The three-seed factorial had suggested a part-SupCon penalty of -6.12 points, $[-10.29, -2.00]$; with five seeds and a later source revision the estimate is a fifth as large and spans zero, so we do not retain it. Factorial diagnostics, including near-collapse of part-triplet embeddings, are in Supplementary Tables S1–S3.

Table 2: CASIA-B development Rank-1 (%), five training seeds per objective, 12 held-out identities and 792 probes. Clean and corrupted-probe columns give mean ± seed standard deviation. Corrupted columns average four families at severities 0.1–0.7; single-view averages NM, BG and CL.

	Global descriptor				Raw parts	
Training objective	Clean	Corr. probe	Corr. gallery	Single-view	Clean	Corr. probe
CE, no masking	80.68 ± 2.79	33.17 ± 0.54	45.53	62.78	83.21	35.49
Masked CE	70.40 ± 3.57	46.92 ± 1.17	56.78	50.45	73.26	50.40
Matched CE	80.33 ± 1.71	58.68 ± 2.59	66.55	65.49	81.77	61.89
Metric baseline	79.60 ± 3.11	60.94 ± 1.07	66.21	65.44	82.35	64.67
Global SupCon	80.76 ± 2.72	62.31 ± 1.86	67.00	66.63	83.43	66.59
Part SupCon	78.94 ± 1.50	59.67 ± 2.72	64.69	64.03	80.83	62.54
Global + part SupCon	79.39 ± 2.29	62.14 ± 1.45	67.14	65.45	81.34	65.09

Table 3: Paired Rank-1 difference from the metric baseline (percentage points; crossed 95% bootstrap interval over seeds and identities), global descriptor. The three-seed column is the earlier factorial on CASIA-B; Gait3D uses its 300-identity development split. Dashes mark objectives absent from a cohort. CE denotes cross-entropy.

(a) CASIA-B development

	Five seeds		Three seeds
Training objective	Clean	Corrupted probe	Corrupted probe
CE, no masking	+1.09 [−3.89, 5.66]	−27.77 [−33.21, −22.29]	–
Masked CE	−9.19 [−14.67, −4.34]	−14.02 [−19.08, −8.98]	–
Matched CE	+0.73 [−3.96, 5.68]	−2.26 [−7.08, 2.86]	–
Global SupCon	+1.16 [−3.46, 5.51]	+1.37 [−2.19, 5.02]	−0.67 [−4.86, 3.79]
Part SupCon	−0.66 [−4.92, 3.66]	−1.27 [−5.24, 2.74]	−6.12 [−10.29, −2.00]
Part triplet	–	–	−3.26 [−7.59, 1.06]
Global + part SupCon	−0.20 [−4.39, 4.12]	+1.21 [−2.54, 4.40]	−3.21 [−8.97, 2.89]
Global + part SupCon, ungated	–	–	−2.19 [−5.35, 1.11]

(b) Gait3D development, three seeds

Training objective	Clean	Corrupted probe	Corrupted gallery
CE, no masking	−7.44 [−12.33, −2.56]	−15.70 [−18.46, −13.01]	−11.81 [−13.87, −9.80]
Matched CE	−4.33 [−8.22, −0.11]	−2.46 [−4.35, −0.39]	−2.15 [−3.46, −0.79]
Global + part SupCon	−1.44 [−5.11, 2.33]	+0.07 [−1.73, 1.97]	+0.44 [−0.82, 1.75]

5.2. Regional score fusion, not a trained head, carries the matching gain

Table 4 freezes the metric-baseline checkpoints and changes only the descriptor. On the five-seed CASIA-B cohort, reliability-weighted raw parts improve clean Rank-1 by 2.80 points, $[0.30, 4.77]$, and corrupted-probe Rank-

1 by 3.73 points, $[1.67, 5.69]$; the three-seed controls give 2.95 and 4.32 points. The saved untrained heads and all ten fresh random heads also help (corrupted-probe gains of 2.02 to 2.49 points), but less than the deterministic raw pools. On Gait3D development data the corrupted-probe gain, 1.85 points, $[0.75, 2.99]$, falls below the 2-point development criterion; the Gait3D test evaluation was planned before this analysis and used the frozen CASIA-B choice.

Table 4: Checkpoint-only descriptor controls on frozen metric-baseline checkpoints at the declared fusion weight $\lambda = 0.5$: (a) Rank-1 (%) and (b) paired differences (points) with crossed 95% intervals. Controls re-encode each checkpoint, so their global and raw-part values differ from Table 2 by at most 0.05 points.

(a) Rank-1 (%)

	CASIA-B, 3 seeds		CASIA-B, 5 seeds		Gait3D, 3 seeds	
Descriptor	Clean	Corr. probe	Clean	Corr. probe	Clean	Corr. probe
Global only	81.52	63.90	79.60	60.93	35.89	23.96
Saved untrained head	83.04	66.31	81.41	63.22	36.33	25.40
Random heads (mean of 10)	83.75	66.50	81.61	63.12	36.88	25.53
Raw parts, uniform	84.43	68.11	82.40	64.55	37.11	25.89
Raw parts, reliability	84.47	68.23	82.40	64.66	36.89	25.81

(b) Paired differences (points)

Contrast	Endpoint	CASIA-B, 3 seeds	CASIA-B, 5 seeds	Gait3D, 3 seeds
Reliability − global	Clean	+2.95 [1.05, 5.30]	+2.80 [0.30, 4.77]	+1.00 [−2.33, 4.67]
	Corr. probe	+4.32 [2.89, 5.71]	+3.73 [1.67, 5.69]	+1.85 [0.75, 2.99]
Reliability − uniform	Clean	+0.04 [−0.42, 0.42]	+0.00 [−0.28, 0.30]	−0.22 [−1.00, 0.44]
	Corr. probe	+0.12 [−0.02, 0.24]	+0.11 [0.01, 0.22]	−0.08 [−0.23, 0.07]

Reliability weighting is not the source of the gain. Relative to uniform

averaging it changes corrupted-probe Rank-1 by +0.11 points on the five-seed CASIA-B cohort, $[0.01, 0.22]$, by +0.12, $[-0.02, 0.24]$, in the three-seed controls and by −0.08, $[-0.23, 0.07]$, on Gait3D; clean differences are within ±0.22 points. The fusion weight matters more and its best value does not transfer (Supplementary Table S5): on CASIA-B the corrupted-probe gain is 2.04, 3.73, 4.78 and 4.27 points at $\lambda = 0.25$, 0.5, 0.75 and 1, whereas on Gait3D it is 1.30, 1.85, −0.09 and −8.89 points. Selecting $\lambda = 0.75$ from CASIA-B validation would have erased the Gait3D gain, which supports fixing λ in advance.

Figure 3 extends the raw matcher to every trained objective. On CASIA-B it improves corrupted-probe Rank-1 under all seven objectives, by 2.32 (CE without masking) to 4.28 points (global SupCon), and on Gait3D development data by 0.90 to 2.25 points, with every interval above zero; clean differences on Gait3D are imprecise. Within our encoder, the gain therefore depends neither on training the part heads nor on the backbone objective, and Figure 4 shows it across families and severities, including the dynamic family held out from training.

5.3. CASIA-B official test

Only the metric baseline was retrained on all 74 training identities, for a fixed 85 epochs (the median selected development epoch, fixed before these runs), with five seeds. On the standard protocol (Table 5), raw reliability-weighted parts raise the NM/BG/CL average from $48.35 \pm 1.85\%$ to $54.04 \pm 2.18\%$, a paired gain of 5.69 points with interval $[5.19, 6.20]$; the gains are 6.50 points for NM, 5.97 for BG and 4.61 for CL, and every seed improves (5.05 to 6.02 points); uniform and reliability-weighted fusion differ by +0.004 points

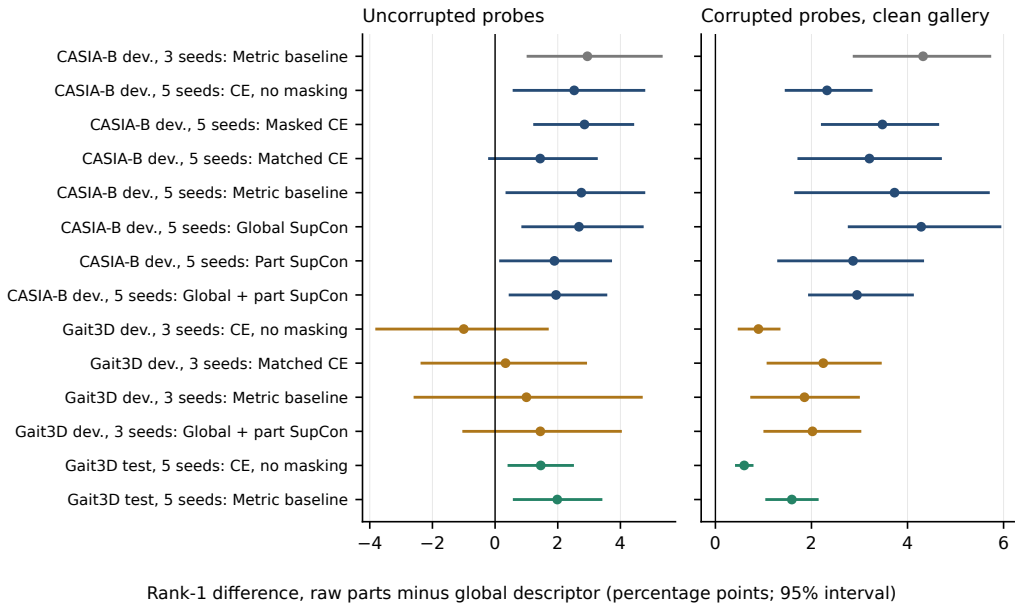


Figure 3: Raw reliability-weighted part matching minus global matching (Rank-1 points, crossed 95% bootstrap intervals) for every cohort and training objective. The corrupted-probe panel averages four families at severities 0.1–0.7. Exact values are listed in Supplementary Table S6.

(Supplementary Table S10 gives the per-condition spread). The absolute level is low: GPGait reproduced on the same split reaches 80.91%, within 0.1 points of its reported 81.01%, and even our GaitGraph2 reproduction (68.17%, 3.66 points below its reported value) exceeds our encoder.

5.4. External evaluation on Gait3D

Table 5 reports the single test evaluation. With the frozen recipe, raw reliability-weighted parts raise the metric baseline's Rank-1 from $21.88 \pm 1.18\%$ to $23.86 \pm 0.68\%$, a paired gain of 1.99 points with crossed interval $[0.61, 3.38]$, and mAP from 16.89% to 17.79%. The gain is also present on

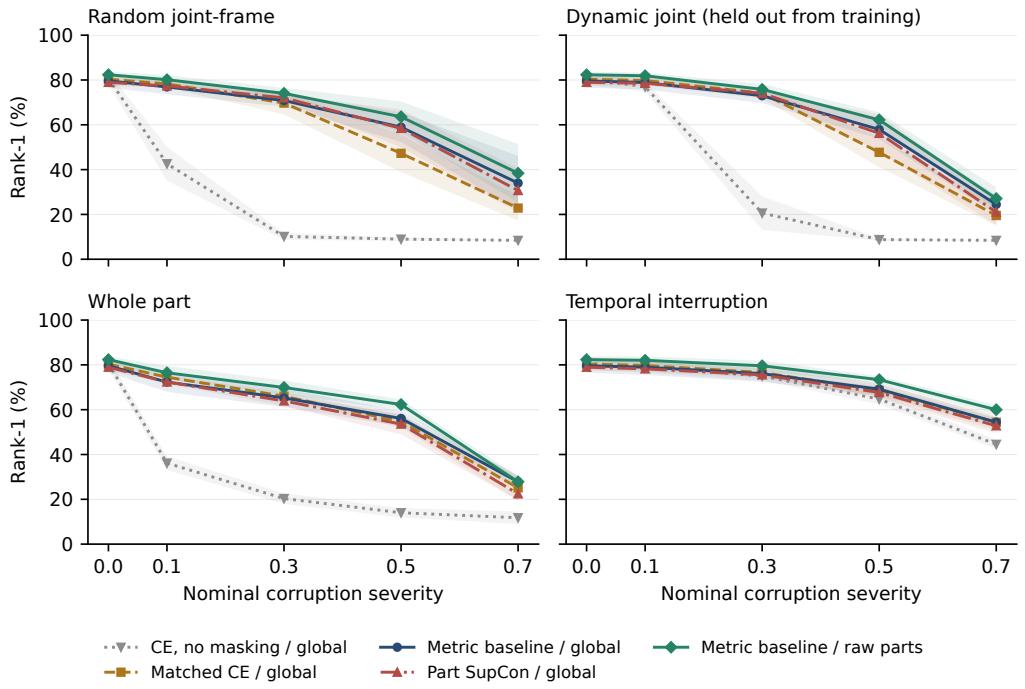


Figure 4: CASIA-B development severity curves with a clean gallery. Lines are means over five training seeds and shaded bands one seed standard deviation. Positive severities average five mask realizations. Severity is nominal and does not imply equal removed fractions across families.

corrupted probes (1.59 points, $[1.07, 2.12]$), with corrupted galleries (1.19, $[0.76, 1.63]$) and for the CE model (1.46 clean points, $[0.44, 2.47]$). For the metric baseline, reliability and uniform weighting coincide to within 0.02 points on every endpoint. The metric baseline exceeds CE by 6.40 clean points, $[4.18, 8.76]$, and 10.62 corrupted-probe points, confirming the development ranking on unseen identities. GPGait trained on the identical subset reaches 23.13% Rank-1 and 16.78% mAP with 18.11 M parameters, against 23.86% and 17.79% for the 2.41 M-parameter encoder with raw parts; our

Table 5: Test-set comparison, Rank-1 and mAP (%). CASIA-B uses identities 075–124 with the standard single-view protocol excluding identical views; Gait3D uses the available-pose subset (947 probes, 5,122 gallery sequences), evaluated once with the recipe frozen on development data. Our rows are means over five seeds of the metric baseline; reproduced baselines were trained once with official FastPoseGait configurations, and reported values are those of the FastPoseGait model zoo [27] on the complete benchmarks. Panel (b) rescores the same GPGait checkpoints (full: all 19 embeddings); raw parts are fused at $\lambda = 0.5$. Parameter counts include identity classifiers. Standard deviations, further metrics and paired intervals are in Supplementary Tables S10–S12.

	CASIA-B official test				Gait3D test		
System	NM	BG	CL	Avg.	Rank-1	mAP	Params. (M) CASIA / G3D
(a) Systems							
Ours, global (5 runs)	67.64	49.50	27.91	48.35	21.88	16.89	1.69 / 2.41
Ours, raw parts (5 runs)	74.14	55.47	32.52	54.04	23.86	17.79	1.69 / 2.41
GaitGraph2, reproduced	76.66	67.23	60.63	68.17	8.87	8.38	0.35 / 0.35
GaitGraph2, reported	80.29	71.40	63.80	71.83	11.20	–	–
GPGait, reproduced	93.77	79.86	69.11	80.91	23.13	16.78	3.65 / 18.11
GPGait, reported	93.60	80.15	69.29	81.01	22.40	–	–
(b) GPGait descriptor variants (same checkpoints)							
Whole-body only	87.86	68.21	59.49	71.85	17.95	12.59	
Whole-body + raw parts	86.53	67.15	59.07	70.92	17.11	11.86	
Full + raw parts	91.55	77.01	67.05	78.53	19.96	14.24	

GaitGraph2 reproduction reaches 8.87%. Because each baseline was trained once and scored by its own evaluator, we read this as parity with GPGait on this subset rather than as an advantage (Rank-5, Rank-10 and mINP are in Supplementary Table S11).

Real detector confidence. Gait3D has no occlusion labels, so we stratified development probes by the fraction of joint-frames with HRNet confidence below 0.5, a proxy fixed before outcomes were examined (Supplementary

Section S7). The pre-specified height-matched contrast of the raw-minus-global gain between the most and least low-confidence tertiles is -0.68 points, $[-7.69, 6.22]$: the gain is not concentrated on low-confidence probes, and these data cannot show that it addresses real occlusion.

5.5. Transfer to GPGait

Table 5(b) applies the matcher to GPGait (paired intervals in Supplementary Table S12). Unlike in our encoder, GPGait's trained part branches carry much of its accuracy: removing them lowers the CASIA-B average by 9.08 points, $[7.43, 10.71]$, and Gait3D Rank-1 by 5.17, $[3.06, 7.29]$. Raw parts neither replace nor complement them: fused with the whole-body descriptor they change the average by -0.94, $[-1.77, -0.15]$, and fused with the full descriptor they lower it by 2.40, $[1.53, 3.29]$, and Gait3D Rank-1 by 3.17, $[1.80, 4.65]$. An exploratory variant chosen after these results, which projects mean-plus-max region pools through each branch's trained whole-body head, also improves neither fusion (Supplementary Table S9). This agrees with GPGait's own ablation, in which removing the partition masks at equal parameter count lowered CASIA-B accuracy from 81.01% to 76.47% and Gait3D Rank-1 from 22.40% to 20.40% [8].

5.6. Cost

The encoder uses 1.54 M parameters at inference and encodes a 48-frame sequence in 4.06 ± 0.07 ms at batch size one on a one-seventh H100 NVL partition (MIG 1g.12gb), excluding gallery matching. Raw parts add no parameters but store 1,536 coordinates and five reliabilities per sequence instead

of 256. The reproduced baselines were not timed, so no speed comparisons are made.

6. Discussion

Which part intervention helps depends on the encoder. In our encoder, auxiliary part supervision did not help and untrained regional pools did; in GPGait, part-specific branches trained with their own losses added 5 to 9 points and raw pools added nothing. The results need not conflict: our part SupCon was an auxiliary loss on shared features, whereas GPGait trains separate branches with restricted connectivity, and raw pools can only add evidence the global embedding lacks. Without both comparisons either intervention could be credited wrongly, so global, raw regional and trained regional matching should be compared on the same frozen checkpoint. Replication also mattered: a part-SupCon penalty supported by three seeds and 12 identities did not survive five seeds.

What the matching gain is and is not. Uniform and reliability-weighted fusion are indistinguishable, and random heads help less than raw pools, so neither confidence gating nor a learned projection is required. Without size-matched random regions the gain is tied to regional score fusion, not to anatomical grouping as such. It is smaller on Gait3D than on CASIA-B and not concentrated on low-confidence probes, so it may reflect a generic ensembling benefit as much as robustness to missing parts.

Absolute performance and silhouettes. With raw parts our encoder matches GPGait on Gait3D but is 27 points below it on CASIA-B, and raw matching does not improve GPGait itself, so the matcher is a cheap remedy for

encoders whose regional features were not trained for retrieval rather than a way to improve strong part-aware systems. Silhouette-based systems remain more accurate still [26, 37, 40, 41], because silhouettes keep body shape that keypoints discard [8]; pose is compact, less affected by clothing and carried objects, and makes body regions explicit. Silhouette models usually train horizontal part strips with their own losses [37, 40]; whether raw regional matching behaves there as in our encoder or as in GPGait is untested.

Outlook. Cross-dataset evaluation, the focus of GPGait [8], would test whether raw regional matching helps across cameras and environments; labelled real occlusion such as the SUSTech1K occlusion condition [36] and larger pose datasets such as OU-MVLP-Pose and GREW [34, 51] would test the occlusion and scale claims; and because the matcher needs no training, it could aid pseudo-labelling of unlabelled gait data by clustering self-supervised contrastive embeddings [19].

7. Limitations

Data and exposure. CASIA-B development comparisons rest on 12 identities, and its test identities were examined in earlier project experiments. Gait3D test identities are unexposed, but cover only the metric baseline and CE on an available-pose subset lacking 53 test identities.

Synthetic and proxy occlusion. Masks applied after normalization omit left–right swaps, cross-person assignments, wrong tracks and occlusion-induced normalization errors, detector confidence is only a proxy for occlusion, and labelled real occlusion was not used.

One encoder and fixed settings. All objectives share one encoder, pose source, loss weights and temperatures, so tuning or other part objectives could change the ranking; random-region controls are absent.

Baselines and statistics. External baselines were trained once and scored by their own evaluators; our GaitGraph2 reproduction fell 2.3 to 3.7 points short. The GPGait test uses clean probes, and removing its part branches removes their architecture and losses together. Many comparisons are exploratory and intervals unadjusted.

Ethical considerations. Gait can identify people without their awareness; robustness gains do not remove the need for a lawful purpose, consent where applicable, access control, retention limits and human oversight, and these datasets support neither fairness nor deployment claims. Operational evaluation should follow established biometric testing principles such as ISO/IEC 19795-1 [52].

8. Conclusion

On our encoder, auxiliary SupCon objectives did not reliably change accuracy, whereas raw part matching from frozen checkpoints improved corrupted-probe Rank-1 under every objective and raised Gait3D test Rank-1 by 1.99 points, $[0.61, 3.38]$, to parity with GPGait; on GPGait, trained part branches added 5 to 9 points and raw matching lowered accuracy. Where a part-aware gain comes from is therefore encoder-dependent, and part-aware systems should report global, raw regional and trained regional matching on the same frozen checkpoints.

CRediT authorship contribution statement

Minas Aslanyan: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Levon Aslanyan:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

CASIA-B poses are the HRNet-derived release distributed through FastPoseGait (`https://github.com/BNU-IVC/FastPoseGait`); the dataset is available from Science Data Bank (`https://www.scidb.cn/en/detail?dataSetId=8ec62efd66a544939e821edeccc1f35c`). Gait3D and its 2D poses are available from the dataset authors (`https://gait3d.github.io/`) under their licence. Neither dataset is redistributed. Identity splits, preprocessing, corruption seeds, per-probe outcomes and the scripts that regenerate every table and figure are available with the code at `https://github.com/likwifi/oap-supcon`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with analysis scripting, consolidation of experimental results and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

[1] R. Liao, S. Yu, W. An, Y. Huang, A model-based gait recognition method with body pose and human prior knowledge, Pattern Recognition 98 (2020) 107069. doi:10.1016/j.patcog.2019.107069.

[2] S. Yan, Y. Xiong, D. Lin, Spatial temporal graph convolutional networks for skeleton-based action recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 32, 2018. doi:10.1609/aaai.v32i1.12328.

[3] T. Teepe, A. Khan, J. Gilg, F. Herzog, S. Hörmann, G. Rigoll, GaitGraph: Graph convolutional network for skeleton-based gait recognition, in: 2021 IEEE International Conference on Image Processing, 2021, pp. 2314–2318. doi:10.1109/ICIP42928.2021.9506717.

[4] T. Teepe, J. Gilg, F. Herzog, S. Hörmann, G. Rigoll, Towards a deeper understanding of skeleton-based gait recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2022, pp. 1569–1577.

[5] T. Li, W. Ma, Y. Zheng, X. Fan, G. Yang, L. Wang, Z. Li, A survey on gait recognition against occlusion: Taxonomy, dataset and methodology, PeerJ Computer Science 10 (2024) e2602. doi:10.7717/peerj-cs.2602.

[6] P. Huang, Y. Peng, S. Hou, C. Cao, X. Liu, Z. He, Y. Huang, Occluded gait recognition with mixture of experts: An action detection perspective, in: Proceedings of the European Conference on Computer Vision, 2024, pp. 380–397. doi:10.1007/978-3-031-72658-3_22.

[7] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, D. Krishnan, Supervised contrastive learning, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 18661–18673.

[8] Y. Fu, S. Meng, S. Hou, X. Hu, Y. Huang, GPGait: Generalized pose-based gait recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 19595–19604.

[9] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 1597–1607.

[10] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, Momentum contrast for unsupervised visual representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 9729–9738.

[11] T. Wang, P. Isola, Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 9929–9939.

[12] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. A. Pires, Z. Guo, M. G. Azar, B. Piot, K. Kavukcuoglu, R. Munos, M. Valko, Bootstrap Your Own Latent - A New Approach to Self-Supervised Learning, in: Advances in Neural Information Processing Systems, Vol. 33, 2020.

[13] X. Chen, K. He, Exploring Simple Siamese Representation Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 15750–15758.

[14] J. Zbontar, L. Jing, I. Misra, Y. LeCun, S. Deny, Barlow Twins: Self-Supervised Learning via Redundancy Reduction, in: Proceedings of the 38th International Conference on Machine Learning, Vol. 139 of Proceedings of Machine Learning Research, 2021, pp. 12310–12320.

[15] A. Bardes, J. Ponce, Y. LeCun, VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning, in: International Conference on Learning Representations, 2022.

[16] F. M. Thoker, H. Doughty, C. G. M. Snoek, Skeleton-contrastive 3d action representation learning, in: Proceedings of the 29th ACM International Conference on Multimedia, 2021, pp. 1655–1663. doi:10.1145/3474085.3475307.

[17] H. Rao, C. Miao, Skeleton prototype contrastive learning with multi-level graph relation modeling for unsupervised person re-identification, in: Quality, Reliability, Security and Robustness in Heterogeneous Systems, Springer, 2024, pp. 306–321. doi:10.1007/978-3-031-65126-7_19.

[18] L. Lin, J. Zhang, J. Liu, Actionlet-Dependent Contrastive Learning for Unsupervised Skeleton-Based Action Recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 2363–2372.

[19] A. Cosma, A. E. Catruna, E. Radoi, On model and data scaling for skeleton-based self-supervised gait recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 40, 2026, pp. 3434–3442. doi:10.1609/aaai.v40i5.37340.

[20] C.-Y. Wu, R. Manmatha, A. J. Smola, P. Krahenbuhl, Sampling Matters in Deep Embedding Learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2017, pp. 2840–2848.

[21] X. Wang, X. Han, W. Huang, D. Dong, M. R. Scott, Multi-Similarity Loss With General Pair Weighting for Deep Metric Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 5022–5030.

[22] K. Musgrave, S. Belongie, S.-N. Lim, A Metric Learning Reality Check, in: Computer Vision – ECCV 2020, 2020, pp. 681–699.

[23] O. Sener, V. Koltun, Multi-Task Learning as Multi-Objective Optimization, in: Advances in Neural Information Processing Systems, Vol. 31, 2018.

[24] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, C. Finn, Gradient Surgery for Multi-Task Learning, in: Advances in Neural Information Processing Systems, Vol. 33, 2020.

[25] C. Fan, J. Ma, D. Jin, C. Shen, S. Yu, SkeletonGait: Gait recognition using skeleton maps, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38, 2024, pp. 1662–1669. doi:10.1609/aaai.v38i2.27933.

[26] C. Fan, J. Liang, C. Shen, S. Hou, Y. Huang, S. Yu, OpenGait: Revisiting gait recognition toward better practicality, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 9707–9716.

[27] S. Meng, Y. Fu, S. Hou, C. Cao, X. Liu, Y. Huang, FastPoseGait: A toolbox and benchmark for efficient pose-based gait recognition, arXiv preprint arXiv:2309.00794 (2023). doi:10.48550/arXiv.2309.00794.

[28] L. Shi, Y. Zhang, J. Cheng, H. Lu, Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 12026–12035.

[29] M. Li, S. Chen, X. Chen, Y. Zhang, Y. Wang, Q. Tian, Actional-Structural Graph Convolutional Networks for Skeleton-Based Action

Recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 3595–3603.

[30] Z. Liu, H. Zhang, Z. Chen, Z. Wang, W. Ouyang, Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 143–152.

[31] K. Cheng, Y. Zhang, X. He, W. Chen, J. Cheng, H. Lu, Skeleton-Based Action Recognition With Shift Graph Convolutional Network, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 183–192.

[32] H. Duan, Y. Zhao, K. Chen, D. Lin, B. Dai, Revisiting Skeleton-Based Action Recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 2969–2978.

[33] Y. Chen, K. Peng, A. Roitberg, D. Schneider, J. Zhang, J. Zheng, Y. Chen, R. Liu, K. Yang, R. Stiefelhagen, Exploring self-supervised skeleton-based action recognition in occluded environments, arXiv preprint arXiv:2309.12029 (2023).

[34] Z. Zhu, X. Guo, T. Yang, J. Huang, J. Deng, G. Huang, D. Du, J. Lu, J. Zhou, Gait recognition in the wild: A benchmark, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 14789–14799.

[35] J. Zheng, X. Liu, W. Liu, L. He, C. Yan, T. Mei, Gait recognition in the wild with dense 3d representations and a benchmark, in: Proceed-

ings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 20228–20237.

[36] C. Shen, C. Fan, W. Wu, R. Wang, G. Q. Huang, S. Yu, LidarGait: Benchmarking 3D Gait Recognition With Point Clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 1054–1063.

[37] H. Chao, Y. He, J. Zhang, J. Feng, GaitSet: Regarding Gait as a Set for Cross-View Gait Recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 33, 2019, pp. 8126–8133. doi:10.1609/aaai.v33i01.33018126.

[38] Z. Zhang, L. Tran, X. Yin, Y. Atoum, X. Liu, J. Wan, N. Wang, Gait Recognition via Disentangled Representation Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 4710–4719.

[39] X. Li, Y. Makihara, C. Xu, Y. Yagi, M. Ren, Gait Recognition via Semi-supervised Disentangled Representation Learning to Identity and Covariate Features, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 13309–13319.

[40] C. Fan, Y. Peng, C. Cao, X. Liu, S. Hou, J. Chi, Y. Huang, Q. Li, Z. He, GaitPart: Temporal part-based model for gait recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 14225–14233.

[41] B. Lin, S. Zhang, X. Yu, Gait recognition via effective global-local feature representation and local temporal aggregation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 14648–14656.

[42] Z. Huang, D. Xue, X. Shen, X. Tian, H. Li, J. Huang, X.-S. Hua, 3d local convolutional neural networks for gait recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 14920–14929.

[43] M. Wang, B. Lin, X. Guo, L. Li, Z. Zhu, J. Sun, S. Zhang, Y. Liu, X. Yu, GaitStrip: Gait Recognition via Effective Strip-based Feature Representations and Multi-Level Framework, in: Proceedings of the Asian Conference on Computer Vision, 2022, pp. 536–551.

[44] X. Huang, D. Zhu, H. Wang, X. Wang, B. Yang, B. He, W. Liu, B. Feng, Context-Sensitive Temporal Feature Learning for Gait Recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 12909–12918.

[45] M. Wang, X. Guo, B. Lin, T. Yang, Z. Zhu, L. Li, S. Zhang, X. Yu, DyGait: Exploiting Dynamic Representations for High-performance Gait Recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 13424–13433.

[46] P. Huang, S. Hou, J. Huang, Y. Huang, Learning A Unified Template for Gait Recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 12459–12469.

[47] A. Hermans, L. Beyer, B. Leibe, In defense of the triplet loss for person re-identification, arXiv preprint arXiv:1703.07737 (2017).

[48] K. Sun, B. Xiao, D. Liu, J. Wang, Deep high-resolution representation learning for human pose estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 5693–5703.

[49] S. Yu, D. Tan, T. Tan, A framework for evaluating the effect of view angle, clothing and carrying condition on gait recognition, in: 18th International Conference on Pattern Recognition, 2006, pp. 441–444. doi:10.1109/ICPR.2006.67.

[50] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C. L. Zitnick, Microsoft COCO: Common objects in context, in: Computer Vision – ECCV 2014, 2014, pp. 740–755. doi:10.1007/978-3-319-10602-1_48.

[51] W. An, S. Yu, Y. Makihara, X. Wu, C. Xu, Y. Yu, R. Liao, Y. Yagi, Performance evaluation of model-based gait on multi-view very large population database with pose sequences, IEEE Transactions on Biometrics, Behavior, and Identity Science 2 (4) (2020) 421–430.

[52] ISO/IEC, Information technology – biometric performance testing and reporting – part 1: Principles and framework, International Standard ISO/IEC 19795-1:2021 (2021).